

I started with a working environment

- The brief asked for a Slack-like app that agents could use and that I could verify.
- I focused on complete workflows, repeatable runs, and evidence that someone else could inspect.

BUILD



Slack workflows

Runnable code and setup

VERIFY



Recorded agent runs

Actions and result checks

EXPLAIN



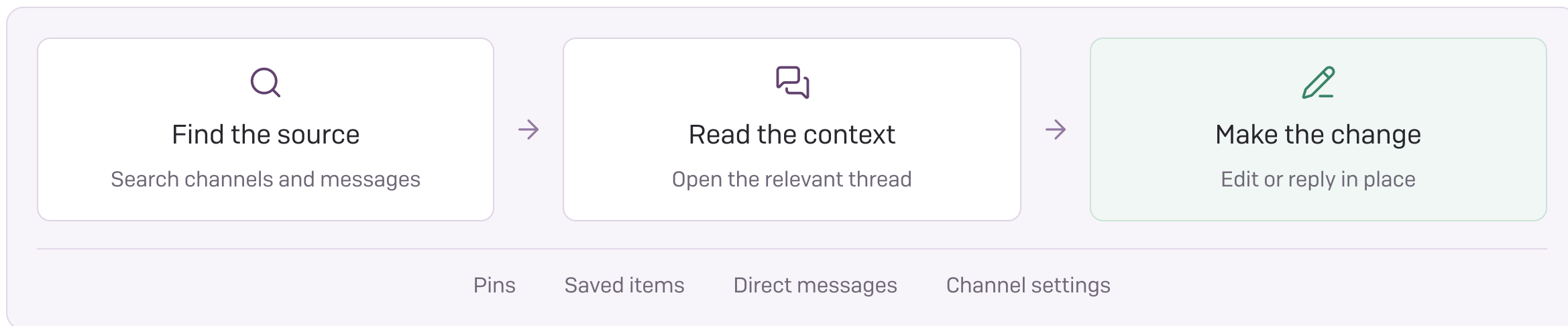
Measured tradeoffs

Isolation, speed, and resources

- Kevin Liu • CUA implementation exercise • October 2026 • Based on the supplied assignment brief.

Build the workflows before expanding the scope

- I built a focused Slack-like app with synthetic people, channels, messages, and decisions.
- Search, threads, and message editing support tasks that require finding information before making a change.
- People and agents use the same workspace UI and state changes.



- [React UI](#) • [SQLite state](#) • [Try the workspace without a key](#) • [Prior-work review](#).

Define success as a change in workspace state

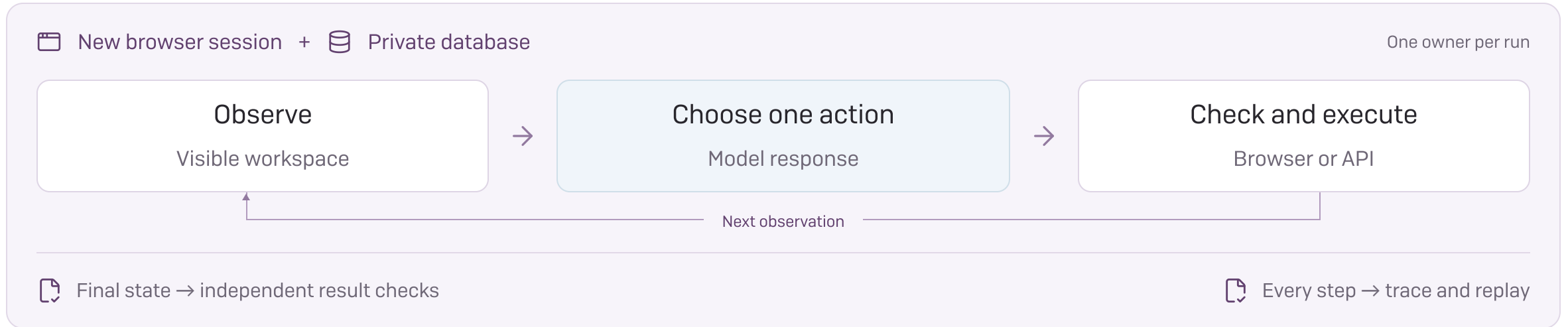
- A plausible action sequence is not enough. The requested result must exist in the right place.
- The checks verify the text, author, destination, and unchanged data.
- The model cannot pass a task by saying that it finished.

 **Example: update a channel topic**
Set the requested topic. Make no other changes.

 Correct topic Exact text in channel settings	 No extra changes Other workspace data stays the same	Both pass Reward = 1
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- Checks grade the final state. A wrong action that the agent fully undoes is not penalized.
- [Task definitions and result checks.](#)

Make each run repeatable and inspectable



- The harness is the code that controls each test run.
 - It resets the workspace, controls the loop, enforces limits, and saves the evidence.
 - Separate workspace data prevents runs from changing each other.
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- Playwright controls the browser. Data isolation is not a security boundary for malicious code.

The harness made a broader comparison possible

- Once I could reset, run, check, and replay a task, I could reuse that process with another model.
- I expanded the project from an environment demonstration into a benchmark prototype.

Original assignment



Working environment

Focused Slack workflows
Known starting state



Reusable harness

Reset · act · check · record
Swap the model



Benchmark prototype

18 tasks × 17 models
One attempt per pair

A repeatable run is necessary. A valid comparison also needs validated tasks.

Separate computer use from direct tool use

Interface	What the agent receives and uses	Earlier passes / attempts
 Pixels	Screenshot → mouse and keyboard	0 / 1
 Accessibility	Labeled controls → element actions	2 / 6
 Page JSON	Structured controls → element actions	2 / 7
 API	Workspace data → direct changes	4 / 6

- The earlier tests used different tasks and models. They do not show which interface is better.
- The 306-run comparison uses accessibility controls only. API results stay separate.

- 20 attempted: 8 passed, 11 incomplete, 1 blocked. 10 planned runs did not start.
- 126 requests · \$0.149874 estimated allowance, including missing-usage reservations · [Earlier interface evidence](#).

Test the software separately from the agents

APP BEHAVIOR



174 backend tests

91 browser tests

Does the software work?

RESULT CHECKS



84 valid states

2,583 invalid states

Does grading reject wrong results?

AGENT BEHAVIOR



306 recorded runs

Can the agent complete the task?

- Scripted browser tests show that the workflows are possible.
- Constructed states test the result checks for the 12 longer tasks.
- Neither test proves that a task instruction is clear to a new reader.

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- [Software verification](#) • Model evidence remains separate from scripted tests.

One attempt for each model and task

$$\begin{array}{ccccccc} \text{🏠} & & \text{☑️} & & \text{🛡️} & & \\ 17 & \times & 18 & \times & 1 & = & 306 \\ \text{model routes} & & \text{tasks} & & \text{attempt per pair} & & \text{recorded attempts} \end{array}$$

- The suite has six simple tasks and 12 tasks that require several changes.
- Every model uses the same task data, action format, and result checks.
- One attempt shows coverage. It does not establish a reliable ranking or repeatability.

40 actions 180 s per run \$5 estimated allowance per run

- 105 earlier attempts retained + 201 continuation attempts. No recorded failure was replaced.
- [Node 24 setup and BYOK instructions](#) · [Exact routes, limits, and run history](#).

Results for 17 models

1 per task/model · Collection complete

306 / 306 attempted · 152 passed · 61 incomplete · 93 blocked

\$189.3110

Usage estimate

\$3.7794

Unresolved · 35 calls

\$193.0904

Total allowance · 306 trials

Model ↕	Trials ↕	Passed ↕	Incomplete ↕	Blocked ↕	Median time ↕	Usage estimate ↕	Unresolved ↕	Total allowance ↕
 GPT-6 Astra	18 / 18	14 / 18	1	3	27.1s	\$35.9623	\$0.0000	\$35.9623
 GPT-6.1 Sol	18 / 18	14 / 18	4	0	28.5s	\$8.4352	\$0.0000	\$8.4352
 GPT-6 Luna	18 / 18	9 / 18	8	1	39.4s	\$0.5394	\$0.0092	\$0.5486
 GPT-OSS 120B	18 / 18	5 / 18	2	11	180.1s	\$0.6682	\$0.1358	\$0.8040
 Claude Fable 5.1	18 / 18	13 / 18	1	4	62.8s	\$48.4714	\$0.0000	\$48.4714
 Claude Opus 5.5	18 / 18	13 / 18	4	1	38.3s	\$28.6987	\$0.3741	\$29.0728
 Claude Sonnet 5.5	18 / 18	7 / 18	4	7	39.9s	\$13.9323	\$0.3304	\$14.2627
 Claude Haiku 4.5	18 / 18	6 / 18	3	9	28.3s	\$3.7921	\$0.0000	\$3.7921
 Grok 4.7	18 / 18	10 / 18	6	2	34.3s	\$11.9101	\$0.3809	\$12.2910
 Qwen 3.8 Max	18 / 18	7 / 18	1	10	125.1s	\$6.9999	\$1.0639	\$8.0638
 DeepSeek V4 Pro	18 / 18	8 / 18	1	9	106.8s	\$4.6816	\$0.2547	\$4.9363
 DeepSeek V4.1 Flash	18 / 18	7 / 18	3	8	65.9s	\$1.3825	\$0.0541	\$1.4365
 GLM 5.3	18 / 18	7 / 18	0	11	125.3s	\$3.6307	\$0.3453	\$3.9760
 GLM 5.3 Flash	18 / 18	8 / 18	0	10	46.2s	\$0.4444	\$0.0000	\$0.4444
 Kimi K3	18 / 18	11 / 18	3	4	50.0s	\$14.0051	\$0.7958	\$14.8009
 MiniMax M3	18 / 18	5 / 18	10	3	25.9s	\$1.4698	\$0.0353	\$1.5052
 Nemotron 3 Ultra	18 / 18	8 / 18	10	0	50.1s	\$4.2874	\$0.0000	\$4.2874

Passed Incomplete Blocked Not run

- 18 public tasks. 1 run per task and model. All use accessibility controls. Median time includes blocked runs.
- Total allowance adds the usage estimate to money reserved for missing usage. These are estimates, not invoices. Including earlier tests: \$194.1058 of the \$300 budget. [Full costs CSV](#) · [JSON](#)
- 105 earlier runs are included. No retries. One run per task cannot establish a reliable model ranking. Two tasks have unclear instructions. [Read the failure analysis](#).
- [Test plan](#) · [Recorded runs](#) · [Download all trials \(CSV\)](#) · [Earlier 36-trial pilot: 20 passed](#)

The checks and the instructions must agree

- I tested the grading code, but did not independently validate every task instruction.
- Trace review found two unclear tasks. Their scores do not cleanly measure model ability.

INSTRUCTION

“DESIGN”

Replacement was not explicit

EXPECTED RESULT

“Willow”

Six completed runs copied DESIGN

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Release sync

Unclear source location
0 / 17 passed

Retrospective

Every run reached a limit
0 / 17 passed

Handoff repair

Sol and Astra clicked Cancel
5 / 17 passed

- Best individual models: 14 / 18 tasks. Across all models: 15 / 18 tasks passed at least once.
- Recorded actions and failure analysis. Keep all original outcomes. A wording fix requires a new task version.

Measure local costs before scaling out



17.34 ms

Create session • median



1.50 ms

Reset state • median

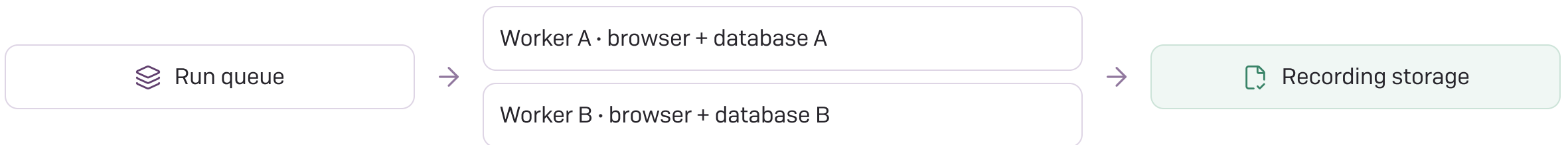


84.15 ms

Screenshot • median

- These timings exclude model calls. They do not measure simultaneous-agent capacity.
- Next, measure full browser memory, sustained throughput, and cleanup under concurrent runs.

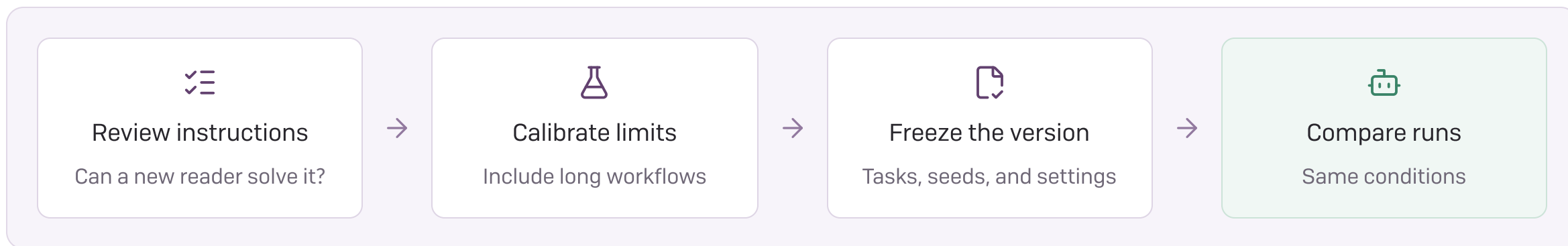
PROPOSED SCALE-OUT



- Apple M5 Max • Node 24 • 100 sessions • 400 reads at concurrency 16 • 10 sequential browser sessions.
- Timing samples and test setup. The multi-worker design is proposed, not measured.

Validate the test before comparing the models

- Have an independent reader solve each task using only its instructions and visible workspace.
- Test limits on long workflows before fixing the comparison settings.
- Use unseen task variants and matched runs to measure reliability and compare interfaces.



The environment is reusable. Benchmark validity requires a separate validation process.

- [Detailed engineering review](#) • [Remaining validation work](#). These are next steps, not completed results.

Inspect the behavior, then discuss the next test



01 · TRY

Use the workspace

Search, open a thread, edit



02 · COMPARE

Choose a model and task

All 306 attempts and costs



03 · REVIEW

Inspect the run

Actions, changes, result checks

- Which workflows best test the behavior you want to improve?
- Should scoring include wrong actions that an agent later undoes?
- What concurrency and cost targets should guide the next version?

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- Relay can reset, execute actions, and return a reward. No reinforcement learning training has been run.
 - [Code and setup](#) · [Requirements and remaining work](#).